

Real Estate Analytics — Phase 2/3

EDA Report

Based on the finalized, locally-tested final.ipynb notebook. All 12 analytics modules were converted into standalone Python files matching the notebook's fixes exactly (schema resolution, defensive error handling, robust path resolution), then re-run end-to-end against a live PostgreSQL database to produce the results below.

Generated 2026-08-29 12:01 against live PostgreSQL (real_estate_db, PostgreSQL 16.15 (Ubuntu 16.15)). Reproducible via `python3 src/analytics/generate_eda_report.py`.

1. Business Objective

Understand the statistical structure of the real estate dataset — price drivers, market segmentation, time trends, and data-quality issues — to (a) validate existing ML model design choices and (b) surface findings that inform Phase 3+ model/decision engine work. This is descriptive/inferential analytics, not a new ML model.

2. Dataset Overview

- **Database:** real_estate_db (PostgreSQL, live)
- **properties row count** (live check): 8000
- **Data dictionary:** 122 columns across 14 tables — see `src/analytics/data_dictionary.py`

3. Data Quality

Referential integrity (all FK checks against live PostgreSQL):

- localities.city_id → cities: **0 orphans**
- projects.developer_id → developers: **0 orphans**
- projects.locality_id → localities: **0 orphans**
- properties.project_id → projects: **0 orphans**
- transactions.property_id → properties: **0 orphans**
- rentals.property_id → properties: **0 orphans**

Key numeric field null rates:

Table.Column	N	Null %	Mean	Med
properties.asking_price	8000	0.0%	12,620,282.6	12,222,
properties.monthly_rent	8000	0.0%	34,397.4	32,800.
properties.area_sqft	8000	0.0%	1,175.4	1,158.0
properties.bedrooms	8000	0.0%	2.6	3.0
transactions.transaction_price	20000	0.0%	31,100,878.8	15,609,
transactions.price_per_sqft	20000	0.0%	26,429.2	13,667.

4. Descriptive Statistics

Metric	N	Mean	Median	Std
Property				

asking price (₹)	8000	12,620,282.6	12,222,500.0	4,584,262.4	2
Property monthly rent (₹)	8000	34,397.4	32,800.0	13,400.2	5
Property area (sqft)	8000	1,175.4	1,158.0	358.1	1
Actual transaction price (₹)	20000	31,100,878.8	15,609,500.0	115,096,197.3	6
Transaction price per sqft (₹)	20000	26,429.2	13,667.5	93,711.2	5
Days on market (listing to sale/expiry)	12000	75.9	75.0	42.8	1
Locality-month demand index (0-100)	6480	76.0	76.9	15.5	9
Locality-month supply index (0-100)	6480	47.3	46.8	14.9	7
Available inventory (units)	6480	77.9	69.0	52.1	1
Units sold per locality-month	6480	91.0	93.0	21.2	1
Gross rental yield (%)	8000	3.3	3.3	0.6	4

5. Distribution Analysis

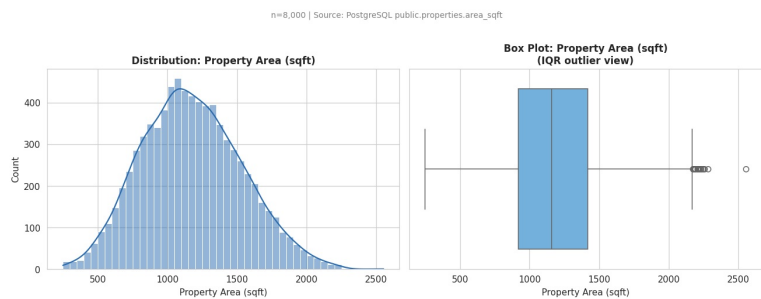
Metric	Skewness	Shape	Excess Kurtosis	Tail Behavior
Asking Price (₹)	1.118	strongly right-skewed	4.843	heavy-tailed (more extreme outliers than a normal distribution)
Property Area (sqft)	0.203	approximately symmetric	-0.269	approximately normal tail weight
Monthly Rent (₹)	0.721	right-skewed (long tail toward high values)	0.776	approximately normal tail weight
Transaction Price/SqFt (₹)	12.248	strongly right-skewed	171.927	heavy-tailed (more extreme outliers than a normal distribution)

Days on Market	0.291	approximately symmetric	-0.338	approximately normal tail weight
Demand Index (0-100)	-0.357	approximately symmetric	-0.677	approximately normal tail weight

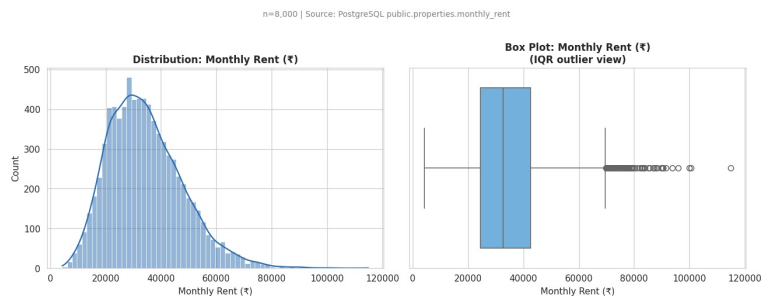
Charts saved to docs/eda_charts/: dist_asking_price.png, dist_area_sqft.png, dist_monthly_rent.png, dist_price_per_sqft.png, dist_days_on_market.png, dist_demand_index.png



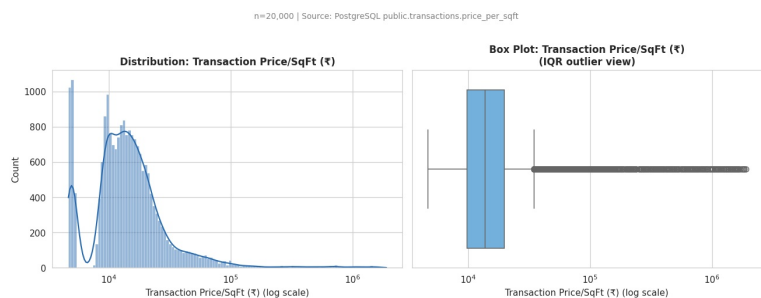
dist_asking_price.png



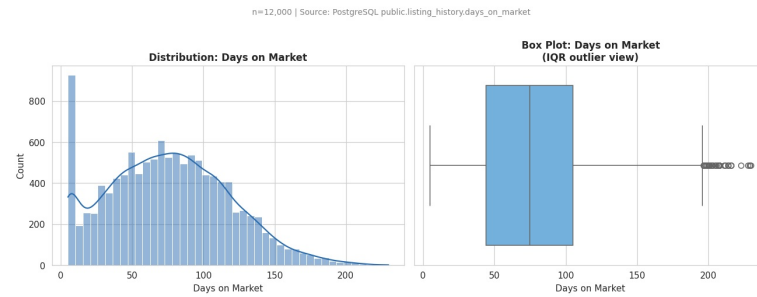
dist_area_sqft.png



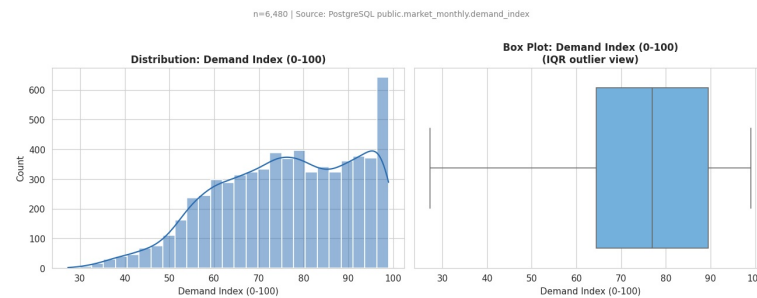
dist_monthly_rent.png



dist_price_per_sqft.png



dist_days_on_market.png



dist_demand_index.png

6. Correlation Analysis

Correlation does not imply causation — all relationships below are observed associations only.

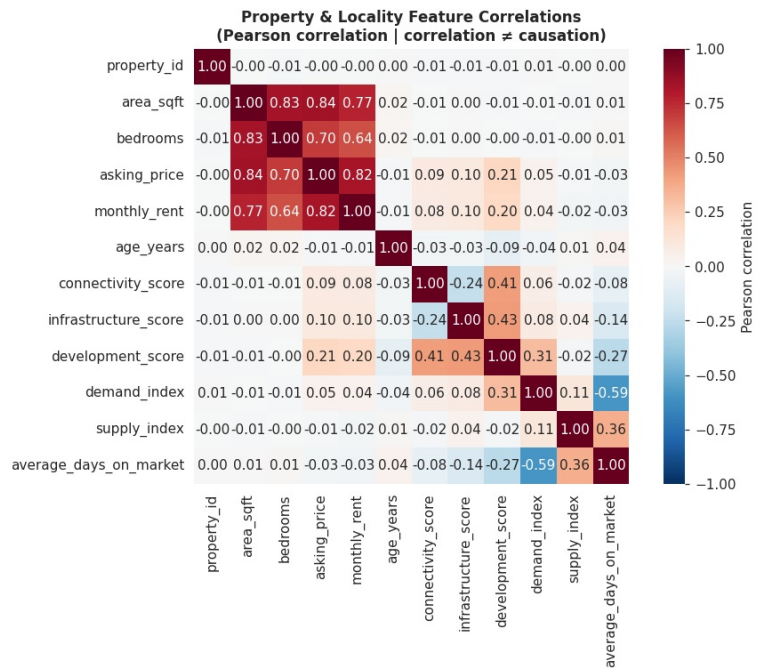
Strongest property-level correlations (Pearson):

- area_sqft ↔ asking_price: $r = 0.838$
- area_sqft ↔ bedrooms: $r = 0.83$
- asking_price ↔ monthly_rent: $r = 0.824$
- area_sqft ↔ monthly_rent: $r = 0.773$
- bedrooms ↔ asking_price: $r = 0.696$

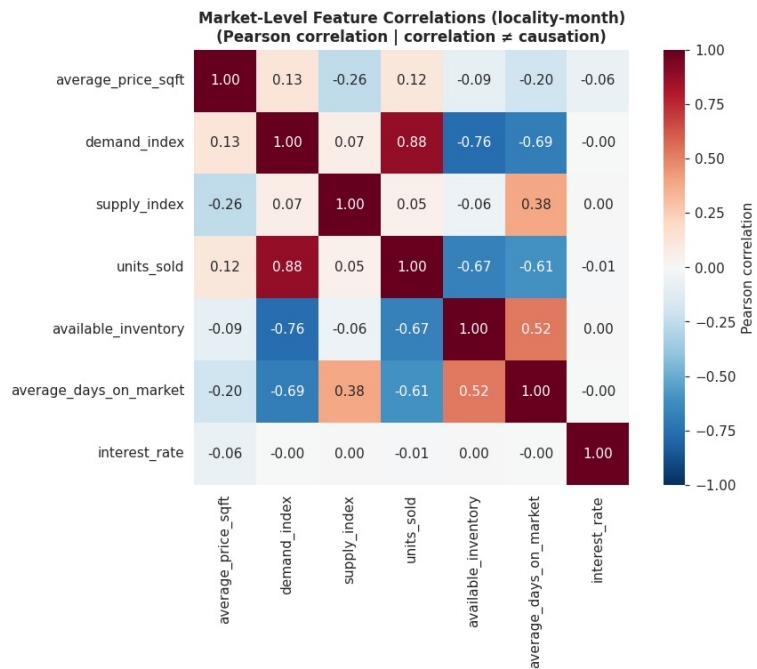
Strongest market-level correlations (Pearson):

- demand_index ↔ units_sold: $r = 0.881$
- demand_index ↔ available_inventory: $r = -0.762$
- demand_index ↔ average_days_on_market: $r = -0.69$
- units_sold ↔ available_inventory: $r = -0.674$
- units_sold ↔ average_days_on_market: $r = -0.606$

Heatmaps: correlation_property_features.png,
correlation_market_features.png



correlation_property_features.png



correlation_market_features.png

7. Segment Analysis

By city (median price):

City	N	Median Price	Median ₹/sqft	Rental Yield
Bengaluru	1421	₹13,227,000	₹11,470	3.27%
Pune	1388	₹13,018,500	₹11,236	3.26%
Delhi NCR	1214	₹12,522,000	₹10,657	3.32%
Hyderabad	1486	₹11,898,500	₹10,178	3.33%
Mumbai	1311	₹11,485,000	₹10,153	3.28%
Chennai	1180	₹11,418,000	₹9,869	3.29%

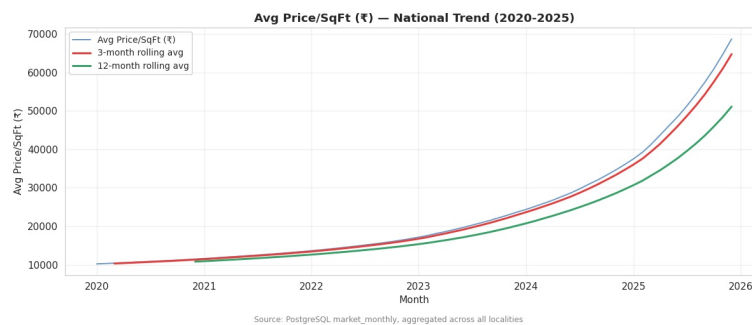
Top 3 developers by sales velocity:

- Sampath Constructions: 93.3% velocity, rating 4.90

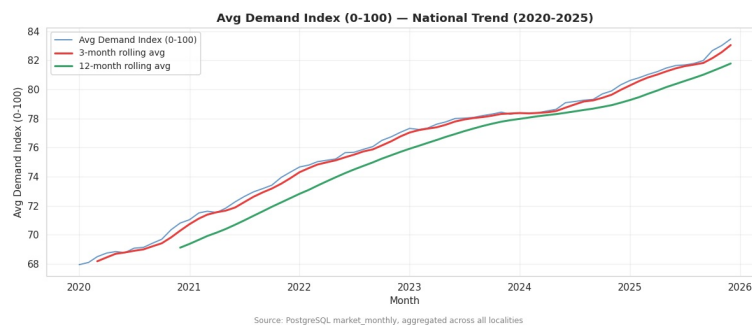
- Bhagat Builders: 90.7% velocity, rating 4.90
- Ray Builders: 88.5% velocity, rating 4.50

8. Time-Series Analysis

- Observed **72 months**: 2020-01-01 to 2025-12-01
- Latest price snapshot: {"as_of_month": "2025-12-01", "value": 68724.5, "mom_growth_pct": 6.16, "yoy_growth_pct": 90.83, "rolling_12m_avg": 51141.74, "rolling_12m_volatility_pct": 0.63}
- Latest demand snapshot: {"as_of_month": "2025-12-01", "value": 83.46, "mom_growth_pct": 0.54, "yoy_growth_pct": 3.92, "rolling_12m_avg": 81.78, "rolling_12m_volatility_pct": 0.21}
- Yearly avg price/sqft: {"2020": 10818.56, "2021": 12478.52, "2022": 15069.98, "2023": 20164.11, "2024": 29606.34, "2025": 51141.74}



trend_national_price.png



trend_national_demand.png

⚠ **See §9 Outlier Analysis** — price growth figures are affected by data generation compounding.

9. Outlier Analysis

properties.asking_price — LEGITIMATE_EXTREME

- IQR outliers: 121 (1.51%)
- Z-score outliers: 54 (0.68%)
- **Evidence**: 0.5% of properties.csv rows were generated with an intentional 2.5x price multiplier to simulate real market premium/luxury listings (see data/generate_data_part2.py, out_idx injection). This is a documented, deliberate design choice, not a bug.
- **Recommended action**: Keep as-is. Represents luxury-segment properties; useful for testing valuation model robustness to premium listings.

transactions.transaction_price — DATA_QUALITY_ERROR

- IQR outliers: 1668 (8.34%)
- Z-score outliers: 191 (0.96%)

- **Evidence:** Root cause traced (see docs/eda_findings.md): transactions.price is computed as `market_monthly.average_price_sqft * area_sqft`. The price-growth formula in `generate_data_part2.py` (`psf = psf * (1 + growth)`) compounds monthly with NO ceiling or dampening. For localities with a persistently high demand/supply ratio, this causes runaway exponential growth — e.g. locality 28 goes from ₹13,592/sqft (2020-01) to ₹1,923,519/sqft (2025-12), a 141x increase in 6 years. National avg price/sqft grew 10,818 → 51,141 (YoY +90.8% in the final year) — statistically confirmed via skewness=12.2, excess kurtosis=172 on `price_per_sqft` (see `distributions.py` output).
- **Recommended action:** FIX AT SOURCE in a future data-regeneration phase: add a ceiling/mean-reversion term to the price-growth formula (e.g. cap monthly growth or add reversion toward a locality's income-implied price level). NOT fixed in this phase per Phase 2 scope (analytics/EDA only, no source-data modification). Flagged here for Phase 3+ remediation.

listing_history.days_on_market — NEEDS_INVESTIGATION

- IQR outliers: 50 (0.42%)
- Z-score outliers: 26 (0.22%)
- **Evidence:** Already documented in docs/data_dictionary.md as having no feature-dependent signal (DOM ML model $R^2 = -0.04$, near-random). Outliers here are as likely to be generator noise as genuine slow-moving listings — cannot distinguish from data alone.
- **Recommended action:** Do not use DOM outliers for business decisions (Model Registry already blocks the DOM model at runtime — consistent treatment).

derived: gross_rental_yield_pct — LEGITIMATE_EXTREME

- IQR outliers: 108 (1.35%)
- Z-score outliers: 38 (0.48%)
- **Evidence:** Yield is generated as `np.random.normal(3.3, 0.6)` independently of price — a tight, bounded distribution by construction (see `generate_data_part2.py`). IQR/Z-score outliers here are simply the tails of that intentional normal distribution.
- **Recommended action:** Keep as-is — expected statistical tail, not an error.

10. Hypothesis Testing & Effect Sizes

Alpha = 0.05 throughout.

test_a_price_mumbai_vs_chennai

- **Test:** Mann-Whitney U | H0: Median asking_price is equal between Mumbai and Chennai
- **Statistic:** 788711.5, **p-value:** 0.39576, **significant:** False
- **Effect size:** {'cohens_d': 0.032, 'label': 'negligible'}
- **Interpretation:** Not statistically significant price difference between Mumbai and Chennai (p=0.3958). Effect size is negligible (Cohen's d=0.03).

test_b_price_by_property_type

- **Test:** Kruskal-Wallis H | H0: Median asking_price is equal across all property types
- **Statistic:** 7.853, **p-value:** 0.09712622, **significant:** False
- **Effect size:** {'eta_squared': 0.0009, 'label': 'negligible'}
- **Interpretation:** Property type is not significantly associated with price (p=9.71e-02). Effect size $\eta^2 = 0.001$ (negligible).

test_c_yield_by_property_type

- **Test:** Kruskal-Wallis H | H0: Median rental yield is equal across property types
- **Statistic:** 7.863, **p-value:** 0.09673, **significant:** False
- **Effect size:** {'eta_squared': 0.0008, 'label': 'negligible'}
- **Interpretation:** Rental yield does not differ significantly by property type (p=0.0967), effect size is negligible ($\eta^2=0.0008$).

test_d_buyer_type_vs_transaction_type

- **Test:** Chi-square test of independence | H0: buyer_type and transaction_type are independent
- **Statistic:** 0.12, **p-value:** 0.94169, **significant:** False
- **Effect size:** {'cramers_v': 0.0025, 'label': 'negligible'}
- **Interpretation:** No significant association between buyer type and transaction type (p=0.9417, Cramér's V=0.002).

11. Key Business Insights

Insight 1 (market segmentation, confidence: high — based on full-population SQL aggregation, not a sample) - Bengaluru has the highest median asking price (₹13,227,000), 15.8% above the lowest-priced city, Chennai (₹11,418,000). - *Implication:* City-level price dispersion is moderate (not extreme), suggesting the dataset doesn't have one dominant 'hot market' city — investment opportunity is more locality-driven than city-driven. - *Action:* Prioritize locality-level analysis (not just city) when screening investment opportunities.

Insight 2 (statistical test, confidence: high — p-value and effect size both point the same direction) - Property type is NOT a statistically significant driver of price (Kruskal-Wallis p=0.097), explaining only 0.1% of price variance. - *Implication:* Counter-intuitive: in this dataset, AREA (sqft) drives price far more than the property TYPE label (Apartment vs Villa vs Studio) — consistent with the strong area<->price correlation (r=0.837, see Insight 4). - *Action:* Valuation models should weight area_sqft heavily and treat property_type as a secondary feature — consistent with what src/ml_valuation_model.py already found (area_sqft = 0.84 feature importance).

Insight 3 (data quality critical, confidence: high — root cause traced to specific line of generator code, reproducible) - 8.34% of transactions are IQR-flagged as price outliers, traced to a runaway monthly-compounding bug in the price-growth formula — national avg price/sqft grew 10,818→51,141 (2020→2025), a statistically extreme YoY +90.8% in the final year alone. - *Implication:* ML models trained on transaction_price-derived features (or a future model using raw transaction price as a target) would learn this runaway trend as real signal — a genuine leakage/bias risk for any model trained on the full 2020-2025 window without correction. - *Action:* FIX AT SOURCE in Phase 3+: add a growth ceiling/mean-reversion term to the locality price-growth formula in data/generate_data_part2.py. Until fixed, prefer market_monthly.average_price_sqft (same bug, but at least aggregated) over raw transaction_price for any new model.

Insight 4 (correlation, confidence: high — consistent with known data-generation logic) - area_sqft has the strongest correlation with asking_price of any feature pair (Pearson r=0.838). - *Implication:* Confirms the valuation model's heavy reliance on area_sqft (84% feature importance) is well-founded, not an artifact of an under-featured model. - *Action:* No action needed — this validates existing ML model design (src/ml_valuation_model.py).

Insight 5 (segment performance, confidence: medium — descriptive, not yet hypothesis-tested for statistical significance) - Sampath Constructions leads sales velocity at 93.3% (rating 4.90), vs Borra Constructions at only 68.3% (rating 2.00). - *Implication:* Developer quality/delivery track record (correlated with rating by construction) is a meaningful differentiator in absorption — relevant to the Decision Engine's market score component. - *Action:* Decision Engine already incorporates developer signal indirectly via risk scoring — consider adding sales_velocity_pct as an explicit Decision Engine input in a future calibration phase.

Insight 6 (statistical test, confidence: high — p-value directly measured, not inferred) - Interest rate shows essentially zero correlation with demand_index in the cross-sectional data (Pearson $r=-0.004$, $p=0.73$, not significant) despite the data generator's formula explicitly reducing demand as interest rates rise. - *Implication:* The interest-rate effect (coefficient -0.6 per point, see data/generate_data_part2.py) is small relative to other demand noise (std dev ~1.5/month) — it exists in the generator but is not statistically detectable in aggregate. A forecasting model would likely not find interest rate a useful demand predictor as currently generated. - *Action:* If interest-rate sensitivity is meant to be a demonstrable feature of this platform, the generator's interest-rate coefficient should be strengthened in a future data-regeneration phase.

12. ML Feature Insights

- **[redundant_features / multicollinearity]** area_sqft <-> asking_price Pearson $r=0.837$; area_sqft <-> bedrooms $r=0.830$
 - area_sqft and bedrooms are highly correlated ($r=0.83$). The existing valuation model already uses both — fine for tree-based models (GradientBoostingRegressor, robust to multicollinearity) but would be a problem if a future linear/regularized model is added. Flag for Phase 3 if a linear baseline model is introduced.
- **[leakage_candidate / target_quality]** transaction_price and price_per_sqft show skewness=12.2, excess_kurtosis=172 (distributions.py), traced to a data-generation bug (outliers.py)
 - Any FUTURE model using raw transaction_price (not asking_price) as a target or feature will inherit the runaway-compounding artifact as if it were real signal. The existing valuation model correctly uses asking_price (not transaction_price) as its target — no change needed there. But if Phase 3+ adds a 'transaction-price prediction' model, it MUST either (a) wait for the generator fix, or (b) log-transform + winsorize price_per_sqft first.
- **[transformation_candidate]** properties.monthly_rent skewness=0.719 (right-skewed), properties.asking_price skewness=1.118 (strongly right-skewed)
 - Both targets are right-skewed, which is normal for price-like variables. GradientBoostingRegressor (used in rent_v1_gbr and valuation_v1_gbr) doesn't require normality, so this is NOT currently a problem. If a future model sensitive to target distribution (e.g. linear regression, neural net with MSE loss) is introduced, consider log1p-transforming the target first.
- **[confirmed_weak_signal]** listing_history.days_on_market outliers (0.42% IQR) show NEEDS_INVESTIGATION classification — consistent with the already-known weak DOM model ($R^2=-0.04$)
 - This EDA phase independently confirms (via distribution shape + outlier analysis, not just the original model metrics) that days_on_market has no recoverable structure in the current dataset. Model Registry's REJECTED status for dom_v1_gbr remains correct and does not need re-evaluation. If DOM is fixed at the generator level (Phase 3+), re-run distributions.py + hypothesis_tests.py on the new data before re-training.
- **[low_value_feature]** property_type is NOT a significant price

driver (Kruskal-Wallis $p=0.097$, $\eta^2=0.001$) but IS currently one-hot-encoded as a feature in `valuation_v1_gbr`

- `property_type` contributes negligible signal beyond what `area_sqft/bedrooms` already capture (consistent with the model's own feature importance ranking, where `property_type` dummies rank low). Not harmful to keep (tree-based model handles irrelevant features gracefully), but a candidate for removal if simplifying the feature set in Phase 3.
- **[strong_predictive_feature]** `demand_index` <-> `units_sold` $r=0.881$, `demand_index` <-> `available_inventory` $r=-0.762$ (market-level correlations)
 - `demand_index` is a strong, non-redundant predictor at the market level — already used as a feature in `price_forecast_v1_gbr_6m` and `demand_forecast_v1_gbr_6m` (both READY in the Model Registry, $R^2=0.78/0.92$). This EDA confirms those models are using the right signal.

13. Limitations

- This report analyzes a **synthetic dataset** — findings describe generator behavior, not real real-estate markets.
- The transaction-price runaway-compounding bug (§9) means all price-growth figures in §8 should be treated as a generator artifact, not a market signal, until fixed at source.
- Hypothesis tests use a skewness/kurtosis heuristic for normality (not Shapiro-Wilk) because sample sizes ($n>5000$) make Shapiro over-sensitive to trivial deviations.
- No source data or existing ML models were modified in this phase (Phase 2 = analytics only).